

744. Find Smallest Letter Greater Than Target

Solved 

Easy

 Topics

 Companies

 Hint

You are given an array of characters `letters` that is sorted in **non-decreasing order**, and a character `target`. There are **at least two different** characters in `letters`.

Return *the smallest character in `letters` that is lexicographically greater than `target`*. If such a character does not exist, return the first character in `letters`.

Example 1:

Input: `letters = ["c","f","j"], target = "a"`

Output: `"c"`

Explanation: The smallest character that is lexicographically greater than 'a' in letters is 'c'.

Example 2:

Input: `letters = ["c","f","j"], target = "c"`

Output: `"f"`

Explanation: The smallest character that is lexicographically greater than 'c' in letters is 'f'.

Example 3:

Input: letters = ["x","x","y","y"], target = "z"

Output: "x"

Explanation: There are no characters in letters that is lexicographically greater than 'z' so we return letters[0].

Constraints:

- `2 <= letters.length <= 104`
- `letters[i]` is a lowercase English letter.
- `letters` is sorted in **non-decreasing** order.
- `letters` contains at least two different characters.
- `target` is a lowercase English letter.

Python:

class Solution:

```
def nextGreatestLetter(self, L: List[str], target: str) -> str:  
    i=bisect_right(L, target)  
    return L[i] if i<len(L) else L[0]
```

JavaScript:

```
var nextGreatestLetter = function(letters, target) {  
    let lo = 0, hi = letters.length;  
    while (lo < hi) {  
        const mid = Math.floor((lo + hi) / 2);  
        if (letters[mid] <= target) {  
            lo = mid + 1;  
        } else {  
            hi = mid;  
        }  
    }  
    return letters[lo % letters.length];  
};
```

Java:

```
class Solution {
```

```
public char nextGreatestLetter(char[] letters, char target) {  
    char res = letters[0];  
    boolean flag = false;  
  
    for(char ch : letters){  
        if(!flag){  
            if(ch > target){  
                res = ch;  
                flag = !flag;  
            }  
        } else {  
            if(ch > target && ch < res)res = ch;  
        }  
    }  
  
    return res;  
}
```